Team A1: Results (concise)

Flexible computing loads, grid resilience, and carbon. General theory + numerical methods.

Model. A regulated utility picks renewable K_R, fossil K_F, and flexible-load K_M capacity. Residual demand $x = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ has CDF $F(x; K_R)$, density $f > 0$, $F_{K_R} \equiv \partial F/\partial K_R > 0$. Shift ratio $g(x) = F_{K_R}(x)/f(x)$.

Distribution-free cutoffs (pinned by the FOCs, independent of shape; $a = K_F^* - K_M^*$):

$$\varphi_1 = \tfrac{C_M}{R-c} = F(a; K_R^*), \quad \varphi_2 = \tfrac{\beta+p-c-C_F-C_M}{\beta+p-R} = F(K_F^*; K_R^*), \quad \varphi^\circ = \tfrac{\beta+p-c-C_F}{\beta+p-c} = F(K_F^\circ; K_R^\circ).$$

1. Optimal contract. Total fossil is minimized at any capacities by the *surplus contract* $d^c(x) = \min\{(-x)_+, K_M\}$: it runs the load only on renewable surplus and burns *zero* fossil for the load.

2. Distribution-free decarbonization. If the load is adopted ($C_M < R\,F(0; K_R^\circ)$) and the renewable problem is single-peaked, then under d^c, $K_R^c \geq K_R^\circ$ and $\mathbb{E}[\text{fossil}^c] \leq \mathbb{E}[\text{fossil}^\circ]$, both strict when adopted, *for every* F (no shape condition). Relative to no load it is a Pareto improvement (profit up, carbon down, blackout held at $1 - \varphi^\circ$).

3. Carbon classification theorem. With $\mu_{\text{low}}^* = \mathbb{E}[x \mid x \in (-K_R^*, a)]$, $\mu_{\text{mid}}^\circ = \mathbb{E}[x \mid x \in (0, K_F^\circ)]$, $q^\circ = F(0; K_R^\circ)$, define

$$\Phi(F) = \underbrace{\tfrac{C_M}{R-c}\,(a - \mu_{\text{low}}^*)}_{\text{capacity substitution}} - \underbrace{\left[K_F^* - K_F^\circ(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ\right]}_{\text{net utilization}}.$$

Then the net saving $\Delta = \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ satisfies $\Delta = \Phi(F)$, so

$$\boxed{\text{carbon falls} \iff \Phi(F) > 0} \qquad \text{(exact, both directions).}$$

4. Impossibility (why Φ needs moments). $\operatorname{sign}\Delta$ is *not* decidable from the cutoffs and capacities alone; both conditional means $\mu_{\text{low}}^*, \mu_{\text{mid}}^\circ$ are load-bearing ($\partial\Phi/\partial\mu_{\text{low}}^* = -\tfrac{C_M}{R-c} < 0$, $\partial\Phi/\partial\mu_{\text{mid}}^\circ = \varphi^\circ - q^\circ > 0$). The carbon sign is irreducibly distributional.

5. Conditional theorems.

- *Per-mean minimality (sharp iff).* At an interior optimum ($\Delta < 0$), μ_{low}^* alone flips $\operatorname{sign}\Phi$ iff $\tfrac{C_M}{R-c}(\mu_{\text{low}}^* + K_R^*) > |\Delta|$ (D_ℓ); symmetrically $(\varphi^\circ - q^\circ)(K_F^\circ - \mu_{\text{mid}}^\circ) > |\Delta|$ for μ_{mid}°.
- *Carbon–profit frontier / leverage.* Budget family $d_\tau = \min\{(-x)_+ + \tau, K_M, (K_F - x)_+\}$ gives the exact reduction $C(\tau) - C(0) = \Phi_{\text{ft}}(\tau)\,(1 - \Lambda(\tau))$. Under Condition M (sufficient, and false at interior optima, where $\Lambda \in [0.14, 0.22]$ yet the conclusion still holds) ($K_R'(\tau) \leq 0 \leq K_F'(\tau)$), G increasing in K_F and decreasing in K_R gives $\Lambda \leq 0 < 1$: surplus-only ($\tau = 0$) is the carbon-optimal member.
- *Carbon boundary.* $\{\Phi = 0\}$ is the boundary; under monotonicity of $\Phi(C_R)$ (Condition S) the carbon-down set is an interval $\{C_R < C_R^\dagger\}$.

6. Regimes. Φ stitches continuously across the small-/large-FCL boundary: at $K_M = K_R$ the surplus term $c\,F(-K_R; K_R) = 0$, so the small-FCL FOC $(R-c)F(K_F - K_M) + c\,F(-K_M) = C_M$ reduces to the large-FCL one. Adoption widens to $\bar{C}_M^b = \bar{C}_M + c\,q^\circ > \bar{C}_M$, $\bar{C}_M = (R-c)\varphi^\circ$; the carbon-down set is the cheap-renewable corner in both regimes.

Verification. $\Phi = \Delta$ to $\sim 10^{-9}$ across the Beta scale-mixture library; per-mean iff, Conditions M/S checked; regime suite 8/8; the solver's Ψ equals Φ with 0 sign mismatches.